Vaibhavee Singh

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Work Experience

Software Development Engineer Intern – Data Platform, Blinkit *April 2026 – October 2026*

- **Planning Tool:** Built GET, PUT, and DELETE REST API endpoints and refactored the capacity planner from static, tenant-specific filter configurations to a dynamic API-driven system with runtime filter management, bulk updates, multi-tenant support, React UI integration, unit tests, and YAML-defined forecast-specific metric ordering.
- **Metric Sync:** Built an automated Airflow DAG to discover and synchronize new business-unit metrics into AWS RDS, streamlining metric onboarding and keeping planning configurations consistent across tenants.
- **ETL Helper Agent:** Built an AI agent for self-service Airflow DAG debugging, integrating Airflow, Commuter, Slack, and Grafana MCPs with custom diagnostics for Spot Instance interruptions and OOM failures; enabled analysts to self-resolve 200+ DAG run failures per day previously handled by the Data Platform team.
- **JupyterHub:** Added support for experimental environments, Claude Code, and GitHub CLI in the internal JupyterHub deployment serving 1,400+ users across the organization.
- **Internal Python Library:** Contributed reusable Jupyter notebook utilities for Apache Pinot data ingestion, segment creation, S3 data management, datatype mapping, and Kafka writes from pre-production environments.
- **AWS Migrations:** Led cross-account AWS migrations of Redash (UI, Redis, RDS) and Commuter; shared KMS-encrypted RDS cluster snapshots and restored them in the target account with minimal downtime, and automated provisioning with Terraform and Atlantis with deployments managed via Argo CD.
- **Airflow & Pinot Automation:** Designed a DAG to automate Apache Pinot table onboarding for Offline and Realtime tables, and developed a custom Apache Airflow plugin using Extra Links to open Commuter output notebooks directly from task instances.
- **Notebook Visualization:** Optimized the Commuter platform with paginated S3 access, 15-minute directory caching with server-side preloading, client/server-side search, and live notebook outputs; upgraded Next.js from 11.1.3 to 16.3.3 and React from 16.8.6 to 18.2.0 to resolve critical security vulnerabilities.

AI Engineering Intern, Scopely *September 2025 – March 2026*

- **MCP Connectors:** Designed and deployed 15+ reusable MCP connectors for Claude Code, Codex, and the internal AI platform, standardizing LLM access to Google Workspace, Confluence, Unity, enterprise applications, internal APIs, and semantic search across teams.
- **Internal AI Platform:** Enhanced the company's internal AI chat platform with authentication, admin UI improvements, and role-based access control (RBAC) for MCPs, enabling secure, team-specific AI access for 3,000+ employees.
- **Talent Acquisition:** Contributed to an internal resume-shortlisting tool using semantic search and embeddings to improve candidate filtering accuracy beyond keyword matching.
- **Evaluation Platform:** Built an AI-powered presentation evaluation platform combining slide analysis, speech transcription, and rule/ML-based scoring to standardize judging across 50+ hackathon teams and 3 judges.
- **API Migration:** Migrated the internal AI Assistant from the OpenAI Assistants API to the Responses API, improving reliability and response consistency.
- **AI Evals & Research:** Designed LLM evaluation metrics and integrated automated evals into production AI systems using Braintrust; ran PoCs on enterprise AI platforms (Gemini Enterprise, AWS Quick Suite) for integration with the existing GenAI stack.

Skills

- **Languages:** Python, C++, JavaScript, SQL
- **AI & ML:** LLMs, AI Agents, MCP (Model Context Protocol), Semantic Search, LLM Evaluation, Braintrust, OpenAI API, Anthropic API
- **Data Engineering:** Apache Airflow, Apache Pinot, Kafka, ETL Pipelines, Redis, Jupyter
- **Cloud & DevOps:** AWS (S3, RDS, KMS), Docker, Kubernetes, Terraform, Atlantis, Argo CD, CI/CD, Grafana, Git
- **Web:** React, Next.js, Node.js, REST APIs

Projects & Achievements

- **PrabhaWatt:** Top 10 Grand Finalist at Luminous Techno-X 2024 for a solar energy management and electricity-saving platform.
- **Research Publications:** Published *Identifying Spam Accounts on Instagram* in IEEE IC3SE-2024 proceedings; co-authored the book chapter *Threat Intelligence and Prediction for DDoS Attacks in Cloud Computing*.

Certifications

- AWS Certified Cloud Practitioner — Amazon Web Services
- Generative AI Fundamentals Accreditation — Databricks Academy

Education

Bennett University — Greater Noida, India
B.Tech in Computer Science and Engineering | CGPA: 8.37/10 — Graduating 2026